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ADDITIONAL MATHEMATICS **4037/13**

Paper 1 **May/June 2010**

2 hours

Additional Materials: Answer Booklet/Paper
Graph paper (2 sheets)

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Mathematical Formulae

1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Binomial Theorem

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \ldots + \binom{n}{r} a^{n-r} b^r + \ldots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!}$.

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1.$$

$$\sec^2 A = 1 + \tan^2 A.$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A.$$

Formulae for ΔABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

$$a^2 = b^2 + c^2 - 2bc \cos A.$$

$$\Delta = \frac{1}{2} bc \sin A.$$

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1 Find the coordinates of the points of intersection of the curve $y^2 + y = 10x - 8x^2$ and the line $y + 4x + 1 = 0$.

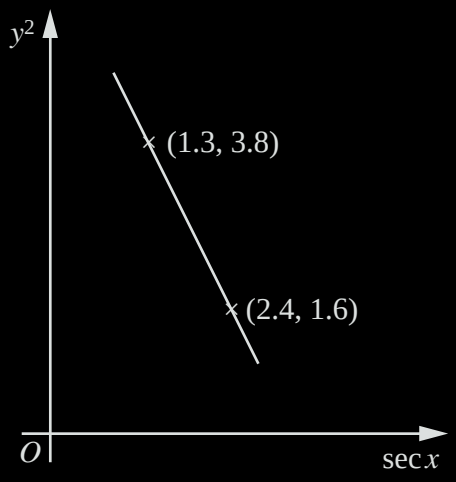
2 The expression $6x^3 + ax^2 - (a + 1)x + b$ has a remainder of 15 when divided by $x + 2$ and a remainder of 24 when divided by $x + 1$. Show that $a = 8$ and find the value of b. [5]

3 Given that $\overrightarrow{OA} = \begin{pmatrix} -17 \\ 25 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$, find

(i) the unit vector parallel to $\overrightarrow{AB}$, [3]

(ii) the vector $\overrightarrow{OC}$, such that $\overrightarrow{AC} = 3\overrightarrow{AB}$. [2]

4



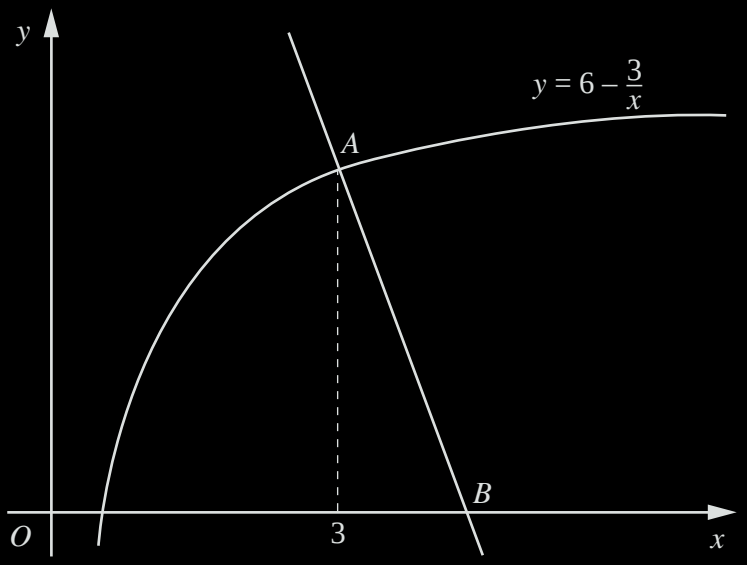
Variables x and y are such that, when y^2 is plotted against $\sec x$, a straight line graph passing through the points (2.4, 1.6) and (1.3, 3.8) is obtained.

(i) Express y^2 in terms of $\sec x$. [3]

(ii) Hence find the exact value of $\cos x$ when $y = 2$. [2]

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5



The diagram shows part of the curve $y = 6 - \frac{3}{x}$ which passes through the point A where $x = 3$. The normal to the curve at the point A meets the x-axis at the point B. Find the coordinates of the point B. [5]

6 **(a)** **(i)** On the same diagram, sketch the curves $y = \cos x$ and $y = 1 + \cos 2x$ for $0 \leqslant x \leqslant 2\pi$. [3]

(ii) Hence state the **number** of solutions of the equation

$$\cos 2x - \cos x + 1 = 0 \text{ where } 0 \leqslant x \leqslant 2\pi.$$ [1]

(b) The function f is given by $f(x) = 5\sin 3x$. Find

(i) the amplitude of f, [1]

(ii) the period of f. [1]

7 The table shows values of the variables p and v which are related by the equation $p = kv^n$, where k and n are constants.

v	10	50	110	230
p	1412	151	53	19

(i) Using graph paper, plot $\lg p$ against $\lg v$ and draw a straight line graph. [3]

Use your graph to estimate

(ii) the value of n, [2]

(iii) the value of p when $v = 170$. [2]

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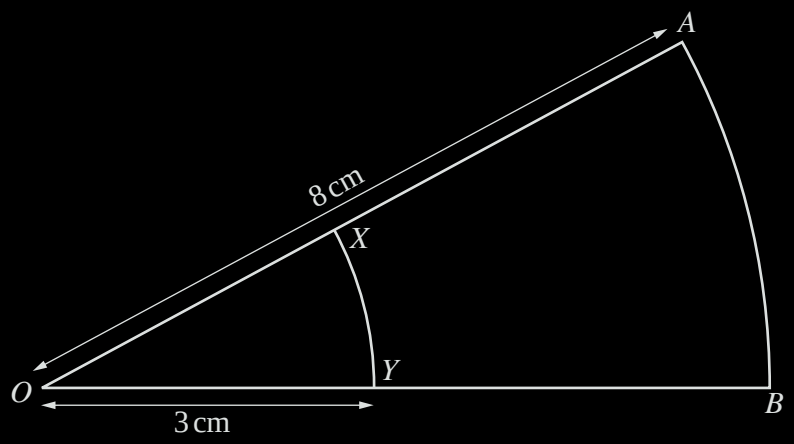
8 Given that $\mathbf{A} = \begin{pmatrix} 4 & 3 \\ 1 & 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} -2 & 0 \\ 1 & 4 \end{pmatrix}$, find

(i) $3\mathbf{A} - 2\mathbf{B}$,

(ii) $\mathbf{A}^{-1}$, [2]

(iii) the matrix $\mathbf{X}$ such that $\mathbf{X}\mathbf{B}^{-1} = \mathbf{A}$. [3]

9



The diagram shows a sector OXY of a circle centre O, radius 3 cm and a sector OAB of a circle centre O, radius 8 cm. The point X lies on the line OA and the point Y lies on the line OB. The perimeter of the region $XABYX$ is 15.5 cm. Find

(i) the angle AOB in radians, [3]

(ii) the ratio of the area of the sector OXY to the area of the region $XABYX$ in the form $p : q$, where p and q are integers. [4]

10 A music student needs to select 7 pieces of music from 6 classical pieces and 4 modern pieces. Find the number of different selections that she can make if

(i) there are no restrictions, [1]

(ii) there are to be only 2 modern pieces included, [2]

(iii) there are to be more classical pieces than modern pieces. [4]

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11 A particle moves in a straight line such that its displacement, x m, from a fixed point O at time t seconds is given by $x = 12\{\ln(2t + 3)\}$. Find

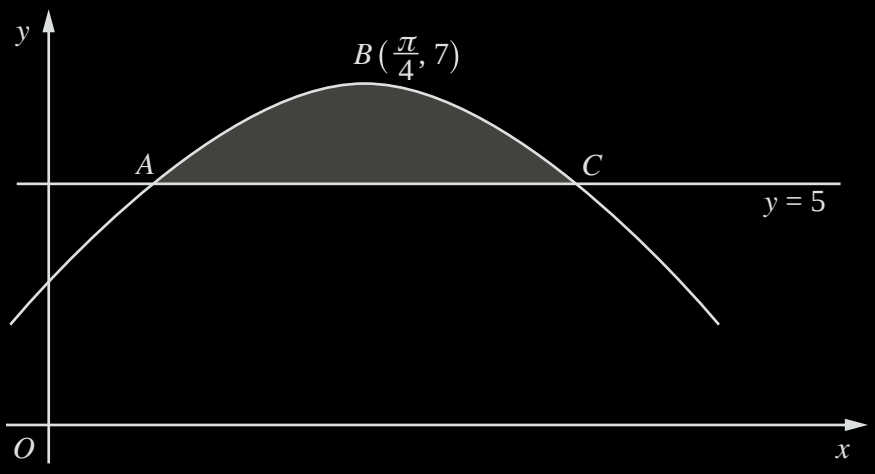
(i) the value of t when the displacement of the particle from O is 48 m,

(ii) the velocity of the particle when $t = 1$, [3]

(iii) the acceleration of the particle when $t = 1$. [3]

12 Answer only **one** of the following two alternatives.

EITHER



The diagram shows part of a curve for which $\frac{dy}{dx} = 8\cos 2x$. The curve passes through the point $B\left(\frac{\pi}{4}, 7\right)$. The line $y = 5$ meets the curve at the points A and C.

(i) Show that the curve has equation $y = 3 + 4\sin 2x$. [3]

(ii) Find the x-coordinate of the point A and of the point C. [4]

(iii) Find the area of the shaded region. [5]

OR

A curve is such that $\frac{dy}{dx} = 6e^{3x} - 12$. The curve passes through the point (0, 1).

(i) Find the equation of the curve. [4]

(ii) Find the coordinates of the stationary point of the curve. [3]

(iii) Determine the nature of the stationary point. [2]

(iv) Find the coordinates of the point where the tangent to the curve at the point (0, 1) meets the x-axis. [3]

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